CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 8: Remainder and Factor Theorems

Every formula and result of the chapter on one sheet

1	Division	Dividend $=$ divisor $\times$ quotient $+$ remainder
2	Remainder on dividing by $(x - a)$	$f(a)$
3	Remainder on dividing by $(x + a)$	$f(-a)$
4	Remainder on dividing by $(ax - b)$	$f\left(\frac{b}{a}\right)$
5	Remainder on dividing by $(ax + b)$	$f\left(-\frac{b}{a}\right)$
6	Factor theorem	$(x - a)$ is a factor of $f(x) \Leftrightarrow f(a) = 0$
7	Factor $(ax + b)$	$f\left(-\frac{b}{a}\right) = 0$
8	Test for the factor $(x - 1)$	Sum of all the coefficients $= 0$
9	Test for the factor $(x + 1)$	Sum of coefficients of even powers $=$ sum of coefficients of odd powers
10	Values to try	Factors of the constant term: $\pm 1, \pm 2, \pm 3, \ldots$
11	Cubic in factor form	$f(x) = (x - a)(\text{quadratic})$, then factorise the quadratic